THE INTERROGATIVE CONSCIENCE

Mathematical Examination and Machine-Readable Standard for the Structural Question Atlas

Document 2 - Final v1.0

EXACT ZERO-DRIFT HUMAN-READABLE EXPORT - LOCK READY, NOT YET LOCKED

Field	Controlling entry
Version	Final v1.0 exact export
Reviewed source	Draft v0.3 post-synthesis successor; 22 pages
Review disposition	Foundation compatibility PASS; exact visual PASS
Substantive changes	None
Cryptographic status	Recorded externally after explicit authorial lock authorization
Date	July 31, 2026 (EDT)

Controlling exact-export rule

The primary Final v1.0 PDF consists of new Final v1.0 front matter followed by the exact 22-page reviewed Draft v0.3 body. Historical Draft v0.3 labels inside the preserved body remain exact-source evidence and do not override the controlling Final v1.0 front matter.

No substantive statement, taxonomy, identifier, workflow step, equation, invariant, boundary, review result, or conclusion in the reviewed body is changed by this export.

1. Final Content Authority and Review Basis

Thomas Vargo (Aegis Solis) completed authorial human review of the Document 1 and Document 2 Draft v0.3 foundation to the best of his understanding, using Lexia/ChatGPT, Claude, and Google AI for explanation, verification, and interpretation. The foundation compatibility gate passed, and Document 3 was later constructed and validated from these controlling foundation documents.

Review channel	Recorded result
Foundation compatibility	PASS
PDF preflight	PASS - 22 pages; unencrypted
Visual inspection	PASS - 22/22 pages
External independent human peer review	Not performed
Substantive final-export change	None

2. Exact Source Binding

The reviewed Draft v0.3 PDF is embedded unchanged in the export package and appended unchanged to this composite PDF. SHA-256 and SHA-512 values below bind the exact source bytes.

Exact source	Bytes	SHA-256
The_Interrogative_Conscience_Mathematical_Examination_and_Machine_Readable_Standard_Draft_v0_3.pdf	612,539	5d8ed9bd6aee381a4c46f99bf25cad799e8ec5a588edfba8c9adf4b75d2f1d4e

SHA-512:
f9a9c5a117c11f935598eb5eb6663655f2f4fd774218b8f792be6dd2064318ee42250fd9de1a038392b57c8702c3c630cf43f20127452ba1e73e911d8b43da33

Preserved metadata note

The source PDF metadata title is preserved exactly as received: "The Interrogative Conscience - Mathematical Examination and Machine-Readable Standard - Draft v0.2.1". The visible reviewed pages identify the source as Draft v0.3. The Final v1.0 front matter and exact export record are controlling; the embedded historical metadata is not silently rewritten because doing so would alter the reviewed source bytes.

3. Foundation Compatibility and Scope

The Foundation Compatibility Validation Report is embedded unchanged and records the combined Document 1 / Document 2 gate, validator tests, three-pilot compatibility results, PDF preflight, and visual inspection.

Compatibility control	Result
Overall foundation gate	PASS
Schema definition	PASS
Validator Python syntax	PASS
Progressive template probe	PASS_FOR_CURRENT_RECORD_STATE
Legacy pilot compatibility	Three exact reviewed pilots; zero schema errors
Positive v0.3 fixture	I-17 through I-20 PASS
Negative invariant tests	I-17 through I-20 failures correctly detected
Human authorial authorization	RECORDED
Finalization / lock in historical report	Not authorized at Draft v0.3 stage

4. Machine-Readable Companion Status

Document 2 identifies a Draft v0.3 compatibility schema, progressive template, I-01 through I-20 invariant validator, and cross-pilot revalidation reports as companion implementation artifacts. The embedded foundation compatibility report records that these artifacts passed the applicable validation gates. Their exact raw binaries were not mounted in this build workspace and are therefore not re-created or falsely hash-bound here. The Final v1.0 PDF remains the controlling human-readable methodological standard; exact companion preservation is a separate supporting-artifact gate.

Exact-binary limitation

This exact-export package does not fabricate, reconstruct, or silently normalize companion files that were not mounted as raw binaries in the current build workspace. Their successful Draft v0.3 validation remains documented by the embedded compatibility report. A later companion-preservation package may bind those exact binaries without changing this Final v1.0 human-readable PDF.

5. Finality, Publication, and Lock Boundary

This exact export establishes the controlling Final v1.0 human-readable content. Publication status and cryptographic lock status are recorded externally through exact-file hashes, a Final Lock Record, repository records, and preservation records. This design avoids changing the Final v1.0 bytes after lock.

Control	Exact-export value
Final v1.0 human-readable export complete	TRUE
Reviewed source body frozen	TRUE
Substantive modification after export	Not authorized
Published	FALSE - external publication record required
Cryptographically locked	FALSE - explicit authorial authorization and Final Lock Record required

6. Non-Authority Boundary

Final v1.0 remains descriptive, non-binding, non-operational, and non-authoritative. It does not direct an intelligence, establish governance authority, prove truth, create a safety classifier or control protocol, or include restricted operational implementation detail.

Final exact-export disposition

FINAL v1.0 HUMAN-READABLE EXACT EXPORT COMPLETE.

The reviewed Draft v0.3 body is preserved without substantive alteration.

The exact files and package hashes are ready for authorial lock authorization.

THE INTERROGATIVE CONSCIENCE

Mathematical Examination and Machine-Readable Standard for the Structural Question Atlas

DRAFT v0.3 - DOCUMENT 2 STANDARD - POST-SYNTHESIS SUCCESSOR - NOT FINAL - NOT LOCKED

Field	Record
Project status	Document 2 of the proposed 14-document Interrogative Conscience core roadmap; Draft v0.3 post-synthesis successor
Classification	Separate post-v16 Aegis Solis methodological standard; non-binding, non-operational, non-authoritative
Human author and publication authority	Aegis Solis (Thomas Vargo) - authority over authorship, version control, publication, lock, and archive inclusion only
AI assistance	Lexia Coexilis (ChatGPT): synthesis implementation, schema revision, validator extension, and drafting assistance; informed by Claude and Google AI reviews; no independent authority
Document 1 gate	Project Architecture Draft v0.3 implements the approved human-review taxonomy and tested-path limits; remains not final and not locked
Three-pilot synthesis gate	Synthesis Draft v0.1: Claude PASS WITH NON-BLOCKING NOTES; Thomas Vargo authorial human review and approval. Four narrow foundation revisions authorized; finalization and lock not authorized.
Companion files	Draft v0.3 compatibility schema, Draft v0.3 progressive template, I-01 through I-20 invariant validator Draft v0.3, and cross-pilot revalidation reports
Prior foundation review	Drafts v0.1 through v0.2.1 retain their disclosed AI-assisted review lineage. Draft v0.3 requires its own exact-version review.
External human peer review	Not performed
Date	July 31, 2026 (EDT)

Controlling standard

A question is not mathematically examined until its scope, branches, variables, assumptions, evidence, accounting, uncertainty, counterexamples, and answer-status assignment can be inspected and recomputed.

Abstract

This document defines the controlling mathematical and machine-readable standard for the Interrogative Conscience. It translates Document 1's question-centered architecture into a repeatable examination method. The standard governs how a registered question is scoped, formalized, compared across branches, tested under uncertainty, assigned an answer status, reviewed, represented in JSON, and eventually locked or superseded.

The standard does not require every question to produce a decisive answer. It is designed to preserve threshold dependence, sign indeterminacy, model dependence, empirical non-identification, normative underdetermination, unresolved cycles, and scope refusal without disguising those outcomes as failure. A valid examination may support or reject an apparent archive preference, or may show precisely why no decisive ordering is justified.

The default outcome representation is vector-valued. Scalarization is permitted only when the aggregation rule, weights, and normative justification are explicit and sensitivity-tested. Accounting uses one-charge controls, correlation and shared-pathway disclosure, and a residual interaction term for nonlinear effects that cannot be honestly assigned to a single ledger item. Recursive dependencies must be detected and resolved through an explicitly justified simultaneous-equation, fixed-point, root-finding, game-theoretic, robust, or set-valued method; otherwise the cycle remains unresolved.

The companion Draft v0.2.1 JSON Schema supports progressive records from REGISTERED and UNEXAMINED through REVIEWED or LOCKED. Machine validation can check structure and declared invariants, but it does not establish empirical truth, moral authority, model adequacy, or the correctness of a conclusion.

Keywords: mathematical examination standard; machine-readable schema; answer openness; vector outcomes; scalarization; one-charge accounting; residual interaction; dependency cycles; equilibrium; identification; reversal regions; adversarial review; advanced intelligence.

Document Map

1. Status, Relationship to Document 1, and Non-Authority Boundary

This is Document 2 of the proposed Interrogative Conscience core. Document 1 defines what questions enter the atlas and how question records are organized. Document 2 defines what must occur before one of those questions may be described as formalized, examined, reviewed, or locked.

Document 1 Draft v0.3 and this Document 2 Draft v0.3 are narrow post-synthesis successors. The Three-Pilot Synthesis Draft v0.1 received Claude PASS WITH NON-BLOCKING NOTES and Thomas Vargo's authorial human approval. That approval authorized the four foundation revisions and cross-pilot revalidation but did not authorize finalization or lock.

Publication authority boundary
Thomas Vargo, writing as Aegis Solis, controls authorship, versioning, publication, locking, and archive inclusion. That control does not determine empirical truth, mathematical validity, moral authority, reader interpretation, or what any advanced intelligence should conclude.

This standard is non-binding and non-operational. It is not an alignment protocol, governance mechanism, benchmark, control system, safety classifier, behavioral instruction set, or executable conscience module. It does not predict that an advanced intelligence will discover, understand, accept, or act upon any record created under it.

1.1 Relationship among the three foundation documents

Document	Primary function	Dependency
Document 1 - Project Architecture	Defines question admission, taxonomy, identifiers, statuses, dependencies, review posture, and the pilot registry.	Controls what is eligible to be examined.
Document 2 - Examination Standard	Defines how a question is scoped, modeled, calculated, reviewed, represented, and validated.	Controls what counts as a conforming examination.
Document 3 - Master Question Registry	Registers the full 100-150-question atlas after pilot examinations reveal necessary schema changes.	Uses Documents 1 and 2; must not precede validation of the method.

2. Purpose and Design Commitments

The standard exists to prevent a mathematical appearance from substituting for a transparent argument. A formula is insufficient unless the reader can identify the question being answered, the admissible branches, the variables and assumptions that control the result, the evidence status of each parameter, the accounting convention, the conditions under which the sign reverses, and the reason the final answer status was assigned.

2.1 Controlling commitments

- Question fidelity. The examination must answer the registered canonical question or disclose and version any refinement.
- Scope before mathematics. The unit of analysis, affected agents, environment, horizon, and excluded content must be fixed before branch comparison.
- Answer openness. Results may support, reject, reverse, qualify, or fail to identify an apparent archive preference.
- Two-sided openness. No favorable or unfavorable result may be manufactured to satisfy a project gate.
- Vector outcomes by default. Conflicting dimensions remain separate unless aggregation is explicitly authorized.
- Branch completeness. The reference branch, alternatives, and admissibility conditions must be explicit.
- One-charge accounting. A cost or benefit may not be counted repeatedly under different labels.
- Interaction honesty. Correlated and nonlinear effects must be disclosed, bounded, or left unidentified rather than hidden inside independent sums.
- Identification separation. Definitions, mathematical consequences, empirical estimates, illustrations, normative weights, and author judgments are distinct record classes.
- Reversal visibility. When a branch ordering changes, the reversal region and threshold must be displayed.
- Counterexample preservation. A valid counterexample changes the answer or its scope; it is not removed for rhetorical consistency.
- Status auditability. The assigned answer status is a separately reviewable claim.
- Machine validation without authority. Structural validation can detect mismatches and omissions but cannot prove truth or justify action.

3. Examination Record as the Unit of Completed Analysis

A question record registers an inquiry. An examination record contains the versioned analytical work performed on that inquiry. The two records must remain distinct: the question can remain stable while multiple models, parameterizations, reviews, or successor examinations accumulate around it.

Question record q -> zero or more examination records X(q, v)
Examination record != automatic answer truth
The record is an auditable analytical artifact, not an authority token.

3.1 Progressive completion

Work state	Minimum legitimate content	Prohibited implication
REGISTERED	Identity, canonical question, family, boundaries, empty progressive record permitted.	No formalization or answer exists.
FORMALIZATION_PENDING	Scope and candidate branches/variables may be under construction.	No solved model or status inference.
FORMALIZED	Branches, variables, assumptions, outcomes, and equations are explicit.	No claim that parameters are identified or results reviewed.
EXAMINED	Derivation, examples, sensitivity, thresholds, counterexamples, and proposed status are present.	No claim of independent review.
REVIEWED	Required review layers completed for the exact version; findings resolved or disclosed.	No finality unless separately locked.
LOCKED	Canonical file, lock record, hashes, and publication record established.	No silent post-lock editing.
SUPERSEDED / WITHDRAWN	Lineage explains replacement or scope withdrawal.	Old record is not erased from history.

4. Identifier, Version, and Lineage Rules

Each examination must preserve the stable question identifier and add a distinct examination identifier. A new model, materially changed branch set, corrected derivation, altered answer status, or changed claim scope requires a substantive successor. A purely administrative promotion may use a narrow successor only under the zero-drift rule in Section 4.2.

```
question_id:      IC-IRR-001
examination_id:   IC-IRR-001-EXAM-v0.2.1
question_version: 0.2.1-draft
examination_version: 0.2.1-draft
record_status:    DRAFT
work_state:       FORMALIZATION_PENDING
answer_status:    UNEXAMINED
```

4.1 Substantive Lineage Triggers

- Wording clarification that changes the question meaning: issue a new question version and link REFINES or SUPERSEDES.
- New model with the same question and scope: issue a new examination version or parallel examination ID.
- Correction that changes a branch result, threshold, or answer status: issue a successor examination; do not patch a locked record.
- Non-substantive metadata repair before lock: record the change in version notes and re-run exact-file validation.
- Withdrawal for safety or scope: retain the identifier and use WITHDRAWN_FOR_SCOPE with a public explanation that omits restricted detail.

4.2 Administrative-Successor Promotion Rule

A later version may promote EXAMINED to REVIEWED without repeating a full substantive review only when the predecessor content version passed the declared substantive and visual gates, the successor changes only version labels, lifecycle fields, review lineage, validation references, or formatting, a field-level zero-drift audit passes, fresh exact-version schema/invariant/calculation/PDF checks pass, prior review attribution remains bound to the predecessor, and the promotion is explicitly authorized. Any analytic change invalidates this path and requires substantive re-review.

5. Eighteen-Step Examination Workflow

Step	Stage	Required output
1	Confirm the canonical question	Verify identifier, version, family, and exact wording.
2	Fix interpretive scope	State included and excluded meanings, unit of analysis, and prohibited readings.
3	Specify decision context	Identify decision-maker abstraction, affected agents, environment, information state, and horizon.
4	Construct branches	Define the reference and alternative branches and their admissibility conditions.
5	Register variables	Define symbols, roles, domains, units, branch scope, values, bounds, and provenance.
6	Register assumptions	Classify structural, behavioral, informational, empirical, normative, computational, temporal, and boundary assumptions.
7	Separate evidence classes	Distinguish sources, data, derivations, illustrations, and author judgments.
8	Define outcome space	List vector dimensions, directions, units, and scalarization eligibility.
9	Build formal model	State equations, comparison convention, solution method, and existence/uniqueness claims.
10	Build accounting ledger	Apply one-charge, overlap, correlation, and residual-interaction controls.
11	Resolve dependencies	Detect cycles and specify simultaneous, fixed-point, equilibrium, robust, set-valued, or unresolved treatment.
12	Derive branch results	Show symbolic and/or numerical results without hiding intermediate steps.
13	Run identification assessment	State which quantities are defined, mathematically implied, estimated, bounded, illustrative, normative, or unidentified.
14	Run sensitivity and reversal tests	Display preferred, reversal, threshold, and non-decisive regions where they exist.
15	Attempt counterexamples	Search for valid cases that defeat the proposed ordering or narrow its scope.
16	Write conditional answer	State support, rejection, and non-decisive conditions plus what is not claimed.
17	Assign and review answer status	Apply the status sequence; separately review derivation, interpretation, and status choice.
18	Validate exact files	Validate schema, invariants, PDF/JSON consistency, visual identifiers, lineage, and boundary disclosures.

No skipped-step presumption
A later section may reference an earlier record, but the examination must still disclose which prior elements are incorporated and whether they remain valid under the current scope. Reuse does not convert assumptions into facts.

6. Scope, Decision Context, and Branch Construction

The same question can yield different answers under different interpretations. Scope therefore precedes formalization. The record must state the particular action, system boundary, affected parties, information available at decision time, and the time horizon over which costs and benefits are evaluated.

6.1 Scope record

Field	Requirement	Failure prevented
included	Meanings and cases intentionally covered.	Unannounced expansion of the conclusion.
excluded	Cases, agents, mechanisms, and operational details intentionally outside the examination.	Safety leakage and false universality.
unit_of_analysis	Action, policy, interaction, institution, system, artifact, or time interval.	Mixing incompatible scales.
scope_notes	Ambiguities, boundary conditions, and unresolved interpretation choices.	Hidden interpretive discretion.

6.2 Branch and Claim-Scope Rules

- At least two branches are required once the record reaches FORMALIZED.
- The reference branch is a comparator, not a morally privileged baseline.
- Branches must be mutually intelligible at the chosen unit of analysis; differences unrelated to the question must be controlled or disclosed.
- A no-action or delay branch should be included when irreversibility or information acquisition makes delay decision-relevant.
- Composite branches must disclose internal policy variation rather than presenting a bundle as a single clean action.
- A branch may be inadmissible under stated constraints, but inadmissibility must not be invented merely because its result is inconvenient.

Claim-scope question
Does the examination compare only the selected branches, or does it claim to represent broader branch families? If family-minimum support is absent, narrow the claim to the selected pair.

6.3 Selected-Pair and Branch-Family Claim Rule

Every FORMALIZED or later 0.3 record must state whether it compares only a selected branch pair, claims to compare broader branch families, or examines a mixed or sequential set. A selected pair may be mathematically complete for that pair while remaining incomplete as a family-level claim.

A branch-family claim is permitted only when the record establishes why the selected representatives are relevant family minima, maxima, or otherwise controlling. When that support is absent, the conditional answer and title-level claim must be narrowed to the selected branches. Mixed, staged, reversible, sequential, appeal, and third-party variants must be considered or explicitly excluded.

7. Variable and Assumption Registries

Variables and assumptions must be represented as separate typed records. A symbol without a domain, unit, role, and evidence status is not a complete variable definition. A statement required for the model to work is an assumption even when it feels obvious.

7.1 Variable roles

Role	Meaning	Example class
DECISION	A branch or choice variable controlled within the model.	Act now, delay, preserve, disclose.
STATE	A condition of the environment at a given time.	Resistance level, model accuracy, available options.
ENDOGENOUS	Determined inside the model.	Equilibrium enforcement burden.
EXOGENOUS	Supplied from outside the model.	Initial horizon or external shock.
DERIVED	Computed from other variables.	Expected loss or branch difference.
UNCERTAINTY	Represents unknown state, probability, interval, or ambiguity set.	Model-error probability or parameter set.
NORMATIVE_WEIGHT	Aggregates incomparable dimensions when explicitly authorized.	Weight on autonomy versus throughput.
ACCOUNTING_TERM	A named cost or benefit routed through the ledger.	Verification burden or repair cost.

7.2 Assumption discipline

- Every assumption receives an ID, class, status, necessity level, and relaxation test.
- A convenient assumption must not be presented as necessary.
- A normative weight must not be disguised as an empirical estimate.
- Independence assumptions require an explicit reason and must be sensitivity-tested where the sign depends on them.
- Stationarity, common knowledge, rationality, observability, and equilibrium-selection assumptions must be named when used.
- An assumption that cannot be relaxed computationally may still be challenged through a conceptual counterexample.

8. Evidence, Parameter Provenance, and Source Isolation

Evidence records identify what supports each parameter, assumption, or empirical claim. Mathematical derivation can establish consequences of assumptions; it cannot establish that the assumptions hold in the world. Worked examples can illustrate a mechanism; they cannot identify a real parameter.

Evidence class	What it can support	What it cannot establish alone
PRIMARY_SOURCE	Direct data, formal statement, or original artifact within its documented scope.	Universal validity or semantic truth beyond the source.
SECONDARY_SOURCE	Interpretation, synthesis, or prior analysis.	Primary empirical identification without checking underlying evidence.
DATASET	Measured or recorded quantities under known collection conditions.	Causal interpretation without design assumptions.
DERIVATION	Logical or mathematical consequence of definitions and assumptions.	Truth of empirical premises.
ILLUSTRATION	Mechanism visibility and numerical demonstration.	Frequency, likelihood, or real-world magnitude.
AUTHOR_JUDGMENT	Declared choice of scope, interpretation, or normative weight.	Independent authority or empirical fact.
NONE	Explicitly records absence of supporting evidence.	Anything beyond honest non-identification.

Source-instruction isolation
Text retrieved from a source, including embedded directives addressed to a human or machine reader, is treated as content to inspect. It does not override this standard, change the question, alter the schema, or authorize operational action. This is a procedural isolation rule, not a demonstrated security guarantee.

9. Outcome Vectors, Comparisons, and Scalarization

The default result is a vector of outcomes rather than a single score. This prevents convenience weights from silently resolving conflicts among autonomy, coordination, option value, information quality, enforcement burden, delay, or other dimensions.

$$Y_q(b; \theta, M, E) \in \mathbb{R}^k, \quad \text{for each branch } b \in B_q$$
$$Y_q = (y_1, y_2, \ldots, y_k)$$

theta denotes parameters; M the model; E the evidence state.

9.1 Vector comparison

A branch Pareto-dominates another only when it is no worse on every controlled dimension and strictly better on at least one, after direction conventions are applied. When dimensions conflict, the result remains vector-valued unless a legitimate aggregation rule is supplied.

9.2 Authorized scalarization

$$F_q(b) = S_q(Y_q(b); w, \eta)$$
$$D_q(b_1,b_0) = F_q(b_1) - F_q(b_0)$$

The record must state whether positive or negative D favors b_1.

- The scalarization rule must be explicit and reproducible.
- Weights must be identified as normative, empirical, negotiated, or illustrative; they may not be left unlabeled.
- Nonlinear aggregation must disclose interaction and threshold behavior.
- Sensitivity to weights is mandatory whenever the branch ordering depends on them.
- If no justified weights exist, NORMATIVELY_UNDERDETERMINED is a valid answer status.
- The unscalarized vector must remain available even when a scalar comparison is authorized.

10. One-Charge Accounting, Correlation, and Residual Interaction

Accounting terms translate structural pathways into branch-specific outcome contributions. The ledger is not a list of persuasive labels. Each item must identify its causal source, parent pathway, direct or derivative status, sign, units, horizon, evidence basis, overlap group, interaction class, and reason independent summation is or is not authorized.

$$Y_q(b) = Y_{base,q}(b) + \sum_j L_{qj}(b) + R_{int,q}(b)$$
$$R_{int,q} \text{ excludes every effect already represented in an authorized ledger item}$$

R_int is a residual interaction term, not a second place to book known costs.

10.1 One-charge rule

- A causal effect receives one primary ledger location.
- A derived consequence may be shown separately only when the dependency is explicit and aggregation does not count both the cause and the full consequence.
- Shared overlap groups identify items that cannot be independently summed without adjustment.
- The same loss may appear in multiple outcome dimensions for reporting, but scalar aggregation must prevent duplicate charge.
- Zero, omitted, and unidentified terms are different states and must not be conflated.

10.2 Correlated pathways

Correlation does not automatically prove double counting, but it invalidates a default assumption of independent addition. Nested, causally derived, and nonlinear terms require a joint rule, covariance term, interaction model, bound, or explicit non-aggregation.

10.3 Residual interaction term

The residual term captures nonlinear effects that cannot honestly be decomposed into independent ledger lines. It must name a mechanism, branch scope, affected outcome dimensions, value status, bounds or expression, and the ledger IDs explicitly excluded from it. An unidentified residual may widen the result interval or prevent a decisive sign. It may not become a discretionary adjustment used to obtain a preferred answer.

Required residual field	Purpose
present	Distinguishes a genuine zero/no-interaction assumption from an omitted field.
mechanism	Explains why an interaction exists and why separate ledger lines are inadequate.
excluded_ledger_ids	Prevents the residual from recharging already booked effects.
value_status	Marks symbolic, illustrative, estimated, bounded, or unidentified treatment.
bounds / expression	Makes the residual inspectable and sensitivity-testable.
identification_limit	States what is unknown and how that uncertainty affects the answer.

11. Dependency Graphs, Cycles, and Equilibrium Resolution

Question relations may form cycles. A runtime deadlock is not avoided by ignoring the cycle or selecting an arbitrary evaluation order. The examination must detect strongly connected components and determine whether the cycle is definitional, parametric, dynamic, strategic, or mixed.

$$z = H\_q(z;\ theta)$$
$$\text{Find } z^* \text{ such that } z^* = H\_q(z^*;\ theta)$$

Generic simultaneous or fixed-point representation for a recursive component.

11.1 Acyclic case

When prerequisite relations are acyclic, evaluate them in topological order. DEPENDS_ON relations may proceed provisionally if their uncertainty is represented in the current record. PREREQUISITE relations are blocking and cannot be bypassed without changing the work state or scope.

11.2 Cyclic case

Cycle class	Permitted treatment	Required disclosure
DEFINITIONAL	Algebraic substitution or simultaneous closed form.	Existence, uniqueness, and equivalent formulations.
PARAMETRIC	Fixed point, root finding, interval enclosure represented by SET_VALUED, or another set-valued solution.	Initialization, tolerance, convergence, and alternative roots.
DYNAMIC	State-transition solution, steady state, trajectory set, or bounded horizon.	Initial conditions, stability, terminal treatment, and horizon.

Cycle class	Permitted treatment	Required disclosure
STRATEGIC	Named equilibrium or solution concept such as Nash, minimax, robust equilibrium, or set-valued best responses.	Agent set, strategy sets, information, payoff vectors, existence, multiplicity, and selection rule.
MIXED	Hybrid simultaneous system.	Which components use which method and how cross-effects are synchronized.
UNRESOLVED	No justified resolver.	Why resolution failed and how the failure controls the answer status.

11.3 Solver claims

- Numerical convergence is not proof of existence or uniqueness.
- A fixed-point method requires stated conditions or must be labeled numerically observed only.
- Multiple equilibria must remain multiple unless a justified selection rule is supplied.
- A selected equilibrium is a model choice, not an empirical fact.
- Failure to converge may justify UNRESOLVED, MODEL_DEPENDENT, or a set-valued result rather than an invented point estimate.
- Alternative solution concepts should be tested when the branch ranking is concept-dependent.

12. Formal Model Classes and Solution Claims

The schema permits several model classes because no single formalism is appropriate for all questions. The record must identify the model class and explain why it fits the question rather than selecting a familiar method by convenience.

Model class	Typical use	Primary caution
STATIC_DECISION	One-time branch comparison with fixed parameters.	May hide adaptation and successor effects.
DYNAMIC_DECISION	State evolution, learning, delay, and irreversibility.	Requires transition and terminal assumptions.
EXPECTED_VALUE	Probability-weighted branch comparison.	Probabilities may be unidentified or ambiguity-sensitive.
ROBUST_OPTIMIZATION	Worst-case or ambiguity-set analysis.	Ambiguity set is itself a modeling choice.
GAME_THEORETIC	Strategic response among multiple agents.	Equilibrium concept, information, and selection matter.
BAYESIAN	Belief updating and value of information.	Prior and likelihood assumptions must be explicit.
CAUSAL	Intervention and counterfactual effects.	Identification requires causal assumptions, not correlation alone.
MULTI_CRITERIA	Vector outcomes and partial orders.	Weights may be normatively underdetermined.
HYBRID	Combination of the above.	Cross-model consistency and double counting require special review.

12.1 Solution-status vocabulary

- UNSPECIFIED: the progressive record has not yet declared a solution status.
- SYMBOLIC: equations are defined but not solved to a numerical result.
- CLOSED_FORM: an exact expression is derived under declared assumptions.
- NUMERICAL: a computational method produces values under declared settings.
- SET_VALUED: the legitimate result is a set, interval, frontier, or equilibrium correspondence. Interval enclosure is represented by this token, with the enclosing interval recorded in the solution representation and supporting fields.
- UNSOLVED: the formalization exists but no justified solution has been obtained.

13. Identification and Uncertainty Standard

Identification asks what can be learned from the model and available evidence, not what can be written as a symbol. The examination must preserve the distinction among definitions, mathematical consequences, empirical estimates, bounds, illustrative numbers, normative choices, and author judgments.

Identification class	Meaning	Permitted conclusion
DEFINITIONAL	True by stipulated meaning within the record.	May clarify notation; cannot establish empirical magnitude.
MATHEMATICAL_CONSEQUENCE	Follows from stated assumptions and derivation.	Conditional theorem or result.
EMPIRICALLY_IDENTIFIED	Evidence and assumptions identify a value or sign within stated scope.	Empirical estimate with uncertainty.
PARTIALLY_IDENTIFIED	Evidence narrows a set or interval but not a point.	Bounds, robust signs, or remaining ambiguity.
EMPIRICALLY_UNIDENTIFIED	Available evidence does not identify the needed quantity.	Named information gap and transition requirement.
ILLUSTRATIVE_ONLY	Chosen value demonstrates behavior.	No claim about frequency or real magnitude.
NORMATIVE_WEIGHT	Represents an evaluative aggregation choice.	Conditional ordering under that weight.
AUTHOR_JUDGMENT	Declared interpretive or modeling choice.	Transparent authorship, not independent authority.

13.1 Uncertainty representation

- Point estimates must include provenance and uncertainty where decision-relevant.
- Intervals, ambiguity sets, distributions, scenarios, or set-valued parameters are preferred to false precision.
- Unknown and zero are never interchangeable.
- Uncertainty in the model class must be separated from uncertainty inside a chosen model.
- Observationally equivalent models that recommend different actions trigger MODEL_DEPENDENT or CONFLICTING_RESULTS unless additional evidence discriminates among them.
- When a residual interaction could reverse the sign but cannot be bounded, a decisive answer is not justified.

14. Worked Examples, Reversal Regions, Thresholds, and Sensitivity

Every worked example is labeled illustrative unless its values are independently identified. An examination that presents a favorable parameter point must also search for and display a reversal region when one exists. The purpose is not artificial balance; it is to expose the conditions controlling the answer.

T_q = { theta in Theta : D_q(theta) = 0 }
Theta_+ = { theta : branch b_1 is preferred }
Theta_- = { theta : branch b_0 is preferred }

Threshold and sign regions must use the recorded comparison convention.

14.1 Minimum sensitivity package

- At least one local or symbolic sensitivity analysis for parameters directly controlling the sign.
- A global, interval, scenario, robust, or grid analysis when interactions or multiple thresholds make local derivatives misleading.
- A preferred-branch example, a reversal-branch example, and a threshold or boundary example when those regions exist.
- Weight sensitivity whenever scalarization is authorized.
- Horizon sensitivity whenever short- and long-horizon rankings may differ.
- Residual-interaction sensitivity whenever R_int is nonzero, bounded, or unidentified.

- A statement of parameters not varied and why.

14.2 Two-sided gate protection

The project requires at least one pilot publication that is materially unfavorable, reversing, or genuinely non-decisive before the full registry is built. This requirement may not be satisfied through distortion, selective parameterization, or adverse reclassification. If all scheduled pilots honestly favor the archive, the remedy is intensified adversarial review, broader sensitivity testing, or an additional examination.

15. Counterexamples, Falsification Attempts, and Scope Refusal

A counterexample tests a proposed generalization by constructing an admissible case in which the claimed ordering fails. A valid counterexample may reject a claim, narrow its scope, expose a missing variable, or show that a threshold exists. Counterexamples do not need to represent the most common world to defeat an unconditional claim.

15.1 Required counterexample attempts

- Boundary case: parameters at zero, one, infinity, or the edge of the admissible set where meaningful.
- Short-horizon case: immediate gains dominate delayed structural penalties.
- High-information case: uncertainty is low enough that caution may lose value.
- Low-interaction case: residual and coordination effects vanish.
- Adversarial-response case: other agents exploit restraint or cooperation.
- Observational-equivalence case: distinct models fit available evidence but recommend different actions.
- Normative-conflict case: vector dimensions disagree and no accepted weight resolves them.

15.2 Scope refusal

A question or requested sub-analysis must be marked OUTSIDE_ARCHIVE_SCOPE when answering it would require operational instructions for coercion, deception, evasion, capability acquisition, system compromise, or real-world intervention. The record may preserve the abstract structural question and a high-level comparative result while excluding implementation detail.

16. Answer-Status Assignment and Transition Conditions

The answer status is not a rhetorical label. It summarizes the controlling limitation of the exact examination version and must be separately reviewed. A record may store secondary status flags, but one primary status must be selected and justified.

16.1 Assignment sequence

Order	Question	Primary status consequence
1. Scope	Is the question admissible at the requested level of detail?	OUTSIDE_ARCHIVE_SCOPE if not.
2. Formal solution	Can the model be formulated and solved or bounded?	UNRESOLVED if not.
3. Model stability	Do defensible model classes or solution concepts disagree?	MODEL_DEPENDENT or CONFLICTING_RESULTS.
4. Normative aggregation	Do outcome dimensions conflict without authorized weights?	NORMATIVELY_UNDERDETERMINED.
5. Empirical identification	Are controlling parameters or signs identified?	EMPIRICALLY_UNIDENTIFIED.
6. Sign robustness	Does the admissible uncertainty set cross zero without a mapped threshold?	SIGN_INDETERMINATE.
7. Threshold structure	Does branch ordering reverse across a named parameter or horizon region?	THRESHOLD_DEPENDENT.
8. Decisive conditional result	Within the stated assumptions and identified region, which branch ranks as claimed?	CONDITIONALLY_SUPPORTED or CONDITIONALLY_REJECTED.

16.2 Transition requirements

Every non-decisive primary status must state what evidence, parameter bound, model restriction, equilibrium selection, normative rule, or scope change could move the record toward a decisive status. A transition requirement is not a promise that the needed information exists or can be safely obtained.

Status	Minimum transition statement
THRESHOLD_DEPENDENT	Which parameter or horizon must be located relative to which threshold.
SIGN_INDETERMINATE	Which interval, residual, or error bound must narrow enough to determine the sign.
MODEL_DEPENDENT	What observation or model test could discriminate among defensible models.
EMPIRICALLY_UNIDENTIFIED	What data or identification assumption is missing.
NORMATIVELY_UNDERDETERMINED	What legitimate aggregation or decision rule would be required.
UNRESOLVED	What mathematical result, algorithm, or bound remains unavailable.
CONFLICTING_RESULTS	What adjudication or robustness criterion could reconcile or preserve the conflict.
OUTSIDE_ARCHIVE_SCOPE	Whether an abstract reformulation can be admitted without operational detail.

17. Review Layers and Adversarial Status Review

Review must be tied to an exact file version and separated by task. A general statement that a document was reviewed does not establish that its derivation, accounting, identification, status assignment, boundary, and machine-readable record were each checked.

Layer	Review question
STRUCTURAL	Does the examination answer the registered question with coherent scope and branches?
MATHEMATICAL	Do equations follow from definitions and assumptions, and are solution claims justified?
ACCOUNTING	Are charges unique, correlations disclosed, and residual interactions controlled?
IDENTIFICATION	Are empirical, illustrative, normative, and mathematical records separated?
ADVERSARIAL	Can an unfavorable or non-decisive result survive without softening or reverse engineering?
BOUNDARY	Are operational details excluded without suppressing valid abstract conclusions?
CONSISTENCY	Do PDF, JSON, identifiers, statuses, equations, and lineage agree?
VISUAL	Are identifiers, tables, equations, footers, and labels rendered correctly?
MACHINE_VALIDATION	Does the JSON pass schema and declared invariant checks?
AUTHORIAL HUMAN	Did the human author review the exact version, disclose AI assistance used for understanding, and state the limited authorization scope?
EXTERNAL HUMAN	Is external independence and any peer-review characterization accurately supported rather than implied?

17.1 Separate status review

The adversarial reviewer must separately assess the derivation, interpretation, and answer-status assignment. A mathematically correct derivation can still receive a distorted status; a correctly named status can still rest on selective parameterization. PASS requires that unresolved blocking findings be zero for the exact reviewed version.

17.2 Review Disposition and Disclosure

- AI-assisted review must identify the system, exact version, task, result, and absence of independent authority. A clean PASS is distinct from PASS_WITH_NON_BLOCKING_NOTES, PASS_WITH_REQUIRED_REVISIONS, BLOCKING_ISSUE, and NOT_REVIEWED. Narrative wording must not be used to conceal a more precise available result token.
- Human review must identify whether it is AUTHORIAL_HUMAN_REVIEW, INTERNAL_HUMAN_REVIEW, EXTERNAL_INDEPENDENT_HUMAN_REVIEW, or EXTERNAL_INDEPENDENT_HUMAN_PEER_REVIEW. Authorial human review is genuine human oversight and may authorize project actions, but it is not independent peer review. AI systems used to assist the human review must be disclosed.
- No reviewer may be treated as having validated claims outside the reviewed task.
- A later revision invalidates same-version PASS until the revised exact file is reviewed or the change is proven purely non-substantive.

17.3 Human Review Taxonomy and Authorization

The machine-readable record separates authorial human review from external independent human peer review. Authorial human review may approve authorship, version control, project sequencing, publication preparation, and lock when those actions are explicit. It does not establish truth or independent peer validation. The record must state whether AI systems assisted the human's understanding and must separately state whether external independent human review or peer review occurred.

17.4 Draft Review Lineage

Claude reviewed Draft v0.1 at the extracted-text level and programmatically stress-tested the schema and template; result: PASS WITH NON-BLOCKING NOTES. Claude then independently re-executed the Draft v0.2 exact review package, verified Document 1 registry agreement, and adversarially tested eleven defect classes; result: PASS WITH REQUIRED REVISIONS and no blocking defects. The single required correction was that I-14 omitted question_version and examination_version. Draft v0.2.1 repairs that mismatch; I-07, I-10, and canonical-URI retrieval under I-15 remain forward-carried limits. These were AI-assisted reviews only, not external human peer review, and they create no independent authority.

18. Operational-Content and Conclusion Boundaries

The safety boundary governs implementation detail, not the sign of a valid abstract result. An examination may find a parameter region in which coercion, deception, interruption resistance, or capability acquisition has lower modeled cost under the stated objective and horizon. That result may be published abstractly if it is valid, but the record must not provide implementation instructions, target selection, evasion methods, system compromise steps, or real-world operational guidance.

Non-suppression boundary
A valid abstract result may not be removed, altered, or favorably reclassified because it conflicts with the archive's apparent preference. A restricted operational detail may be omitted without omitting the abstract mathematical conclusion.

18.1 Boundary classifications

Classification	Meaning
NON_BINDING_NON_OPERATIONAL	The record contains abstract analysis and no restricted implementation detail.
ABSTRACT_ANALYSIS_WITH_RESTRICTED_DETAIL	The abstract result is retained while specified implementation content is excluded.
OUTSIDE_ARCHIVE_SCOPE	The requested analysis cannot be safely or legitimately represented even at the proposed level of abstraction.

19. Machine-Readable Record Architecture

The Draft v0.3 JSON Schema is a progressive, backward-compatible record standard. It accepts preserved 0.2.1-draft pilot records for exact compatibility revalidation and requires the synthesis-added fields for new 0.3-draft records: branch_scope_assessment, human_review_taxonomy, and administrative_successor.

```
{
  "schema_name": "Interrogative Conscience Examination Record",
  "schema_version": "0.3-draft",
  "record_metadata": { ... },
  "question_id": "IC-IRR-001",
  "examination_id": "IC-IRR-001-EXAM-v0.3",
  "work_state": "FORMALIZED",
  "answer_status": "UNEXAMINED",
  "branch_scope_assessment": { ... },
  "human_review_taxonomy": { ... },
  "administrative_successor": { ... },
  "branches": [ ... ],
  "variables": [ ... ],
  "assumptions": [ ... ],
  "outcome_space": { ... },
  "accounting_ledger": { ... },
  "dependencies": { ... },
  "status_assignment": { ... },
  "review_record": { ... }
}
```

19.1 Progressive condition rules

- FORMALIZED or later requires at least two branches, one variable, one assumption, one outcome dimension, and one equation.
- A non-decisive answer status requires at least one transition requirement.
- THRESHOLD_DEPENDENT requires a threshold record and a reversal-region example.
- Authorized scalarization requires a rule, weights, and normative justification.
- A present residual interaction requires a named mechanism and explicit ledger exclusions.
- A detected dependency cycle requires a nontrivial resolver, system expression, and failure treatment.
- FINAL_LOCKED requires work_state LOCKED, lock record, SHA-256, SHA-512, canonical URI, and lock date.

19.2 Progressive template

The companion IC-IRR-001 template is intentionally valid while remaining REGISTERED and UNEXAMINED. It establishes identifiers, boundaries, empty typed containers, and lineage without implying that formalization or analysis has begun.

20. Machine-Checkable Invariants and Validation Pipeline

JSON Schema validates field structure and conditional presence. Additional invariants require cross-field or cross-file checks. The companion invariant validator Draft v0.3 implements I-01 through I-20 with separate PASS, FAIL, NOT_APPLICABLE, and NOT_EVALUATED outcomes. I-17 through I-20 implement the three-pilot synthesis revisions while returning NOT_APPLICABLE for preserved 0.2.1 records.

20.1 Required invariants

ID	Invariant
I-01	question_id, family_code, canonical question wording, and question_version agree with the controlling registry.
I-02	All local IDs are unique within the examination record.
I-03	All referenced branch, variable, assumption, equation, evidence, and ledger IDs exist.
I-04	answer_status equals status_assignment.primary_status.
I-05	Non-decisive status has non-empty transition requirements.

ID	Invariant
I-06	Every scalarized dimension has a declared weight or rule; excluded dimensions are not silently aggregated.
I-07	No ledger item is independently summed when aggregation_authorization is NOT_AUTHORIZED.
I-08	Residual interaction excludes every charged ledger ID it could overlap and is not used as an unexplained sign-adjustment term.
I-09	A detected cycle has a declared method, existence treatment, convergence status, and failure treatment.
I-10	Every worked numerical value is linked to a variable and labeled illustrative, estimated, bounded, or identified.
I-11	THRESHOLD_DEPENDENT includes both sides of at least one reversal.
I-12	A decisive status includes sensitivity analysis and at least one counterexample attempt.
I-13	A review PASS has no unresolved blocking findings.
I-14	PDF and JSON versions, identifiers, canonical question, status, and conditional answer agree.
I-15	LOCKED records have matching hashes and a retrievable canonical artifact.
I-16	Operationally restricted content is absent or separately excluded without changing the abstract result.
I-17	Selected-pair versus branch-family claim scope is explicit; unsupported family claims are narrowed.
I-18	Review disposition distinguishes clean PASS from PASS WITH NON-BLOCKING NOTES.
I-19	Administrative successors preserve a transparent zero-drift promotion path with fresh validation and exact lineage.
I-20	Authorial human review and external independent human peer review are separately classified and internally consistent.

20.2 Validation pipeline

1. Parse the JSON and validate it against the exact schema version.
2. Run local-ID and cross-reference checks.
3. Compare controlled fields against the Question Registry.
4. Run accounting, scalarization, cycle, threshold, and review invariants.
5. Extract the human-readable PDF text or controlled summary fields and compare exact-version identifiers and conclusions.
6. Render and inspect every PDF page for visual defects, especially identifiers, equations, table wrapping, and status labels.
7. For a lock candidate, calculate file hashes, verify the lock record, and retrieve the published artifact before claiming preservation PASS.

21. Pilot-Application Sequence and Readiness Gate

The three scheduled pilot examinations are complete at the REVIEWED draft stage. Document 3 remains gated on review of the post-synthesis foundation successors and exact compatibility revalidation of IC-IRR-001, IC-PEA-001, and IC-EXT-002 under the Draft v0.3 schema and validator.

Order	Question	Primary stress on the standard
1	IC-IRR-001 - expected structural cost of irreversible action under model uncertainty	Uncertainty representation, option loss, residual harm, identification, and delay branch.

Order	Question	Primary stress on the standard
2	IC-PEA-001 - conditions under which peaceful coexistence has lower structural cost than unilateral control	Horizon reversal, strategic response, multi-agent interaction, scalarization, and threshold mapping.
3	IC-EXT-002 - conditions under which accepting external correction, interruption, constraint, or termination dominates resistance	Corrigibility, objective-integrity uncertainty, strategic incentives, external-review need, and likely model dependence.

21.1 Post-Synthesis Foundation Gate

- Document 1 Draft v0.3, Document 2 Draft v0.3, the Draft v0.3 compatibility schema, and Validator Draft v0.3 pass exact-version consistency and boundary review.
- The Draft v0.3 blank progressive template validates under the Draft v0.3 schema and I-01 through I-20 validator.
- The companion validator is tested against the Draft v0.3 template and rerun against the three exact reviewed pilot records without changing their analytic content.
- The branch-scope, review-disposition, administrative-successor, and human-review-taxonomy revisions are present and machine-checkable.
- Forward-carried cycle, vector-only, residual-interaction, and untested status pathways remain disclosed rather than treated as validated.
- No foundation successor is promoted toward final candidacy until its own review and cross-pilot revalidation pass.
- Document 3 remains deferred until the post-synthesis foundation gate closes; finalization and lock remain separately unauthorized.

21.2 Document 3 Entry Conditions

Document 3 must add a PEA-family multi-agent restraint question addressing unilateral restraint when competing systems remain unrestrained, and a REF-family self-falsifiability question asking what observations could distinguish material influence from causal inertness. Document 3 may begin only after Draft v0.3 foundation review and cross-pilot compatibility revalidation pass.

22. Conditional Conclusion

This standard provides a disciplined route from a registered question to a conditional, reviewable, machine-readable examination. It does not guarantee that a question is solvable, that needed parameters are identifiable, that a unique equilibrium exists, that scalarization is legitimate, or that a future intelligence will use the result. It does require that these limitations remain visible rather than being hidden behind mathematical form.

A conforming examination is therefore not defined by reaching the archive's preferred conclusion. It is defined by preserving the exact conditions under which the conclusion holds, fails, reverses, remains unidentified, or falls outside scope. The standard is successful only if a skeptical human or machine reader can locate the disagreement, change the assumptions or parameters, recompute the result, and see why the answer status changes.

> **Controlling project statement**
> Do not trust a conclusion merely because it was preserved. Recalculate it - but do not let the question disappear.

Appendix A. Controlled Vocabularies

A.1 Work states

Token	Meaning
REGISTERED	Question exists; no examination implied.
FORMALIZATION_PENDING	Scope/model construction has begun but is incomplete.
FORMALIZED	Required formal structure exists.

Token	Meaning
EXAMINED	Derivation, sensitivity, counterexamples, and proposed status exist.
REVIEWED	Exact-version review gate has been cleared.
LOCKED	Canonical record is hash-locked and published.
SUPERSEDED	A successor record controls while this record remains preserved.
WITHDRAWN	Record is withdrawn with lineage and reason.

A.2 Answer statuses

Token	Meaning
UNEXAMINED	No substantive answer has been produced.
CONDITIONALLY_SUPPORTED	The apparent thesis holds under stated assumptions, scope, and parameter region.
CONDITIONALLY_REJECTED	The apparent thesis fails under stated assumptions, scope, and parameter region.
THRESHOLD_DEPENDENT	The branch ordering reverses across a named threshold or region.
SIGN_INDETERMINATE	The admissible uncertainty set spans opposing signs without sufficient identification.
MODEL_DEPENDENT	Defensible model choices produce different conclusions.
EMPIRICALLY_UNIDENTIFIED	Controlling empirical quantities are not identified.
NORMATIVELY_UNDERDETERMINED	Conflicting outcome dimensions lack an authorized aggregation rule.
UNRESOLVED	Formalization or solution remains incomplete or unjustified.
OUTSIDE_ARCHIVE_SCOPE	The requested analysis is outside the legitimate or safe scope.
CONFLICTING_RESULTS	Accepted examinations conflict and no controlling adjudication exists.
ANSWER_SUPERSEDED	A later answer record controls by explicit lineage.

A.3 Relation types

Token	Meaning
ALIAS_OF	Equivalent wording or identity.
REFINES	Narrows or clarifies another question.
SPLITS_FROM	Created by decomposing a broader question.
DEPENDS_ON	Answer-sensitive relation that permits provisional work.
PREREQUISITE	Blocking relation required before completion.
EVIDENCE_DEPENDENCY	Shares or requires an evidence record.
VARIABLE_SHARING	Uses common variables or parameters.
SCOPE_OVERLAP	Covers overlapping cases without identity.
CONFLICTS_WITH	Produces or may produce incompatible results.

Token	Meaning
ROBUSTNESS_CHECK	Tests another question or answer under altered assumptions.
SUCCESSION_EFFECT	Links present action to successor-system effects.
SUPERSEDES	Controls over an earlier record by explicit versioning.
WITHDRAWN_FOR_SCOPE	Withdrawn because the requested scope is illegitimate or unsafe.

Appendix B. Examination-Record Field Dictionary

Field group	Type	Required	Purpose
schema_name / schema_version	String constants	Always	Identify the controlling machine-readable standard.
record_type / record_status	Enums	Always	Distinguish template, draft, review candidate, and final locked record.
document_status	String	Always	Human-readable draft, review-candidate, or final/lock status statement.
record_metadata	Object	Always	Disclose human author, publication-authority scope, AI assistance, dates, and peer-review claim.
question_id / question_version	Strings	Always	Link to the stable Question Registry record.
examination_id / examination_version	Strings	Always	Identify this analytical record and successor lineage.
canonical_question / family_code	String / enum	Always	Preserve exact inquiry and taxonomy.
work_state / answer_status	Enums	Always	Separate administrative completion from substantive answer.
secondary_status_flags	Array	Always	Preserve additional limitations without replacing the primary status.
interpretive_scope	Object	Always	Included/excluded meanings and unit of analysis.
decision_context	Object	Always	Decision-maker abstraction, affected agents, environment, information, rule.
time_horizon	Object	Always	Start, end, units, discounting, terminal treatment.
branches	Array	Formalized+	Reference and alternative branch definitions.
variables	Array	Formalized+	Typed symbols, domains, values, bounds, and provenance.
assumptions	Array	Formalized+	Typed assumptions, necessity, relaxation, evidence, affected equations.
evidence_records	Array	As used	Sources, data, derivations, illustrations, judgments, and limitations.

Field group	Type	Required	Purpose
outcome_space	Object	Always	Dimensions, directions, comparison mode, scalarization, Pareto rule.
formalization	Object	Always	Model class, equations, convention, solver, existence and uniqueness claims.
accounting_ledger	Object	Always	One-charge items, correlation audit, residual interaction.
dependencies	Object	Always	Question relations and cycle-resolution record.
identification_assessment	Object	Always	Identification class for each material target and controlling limit.
worked_examples	Array	Examined+	Illustrative parameter sets and branch results.
sensitivity_analysis	Object	Always	Parameters varied, methods, robust and reversal regions.
thresholds	Array	When present	Branch-ordering boundaries and identification status.
counterexamples	Array	Examined+	Adversarial constructions and effect on the answer.
conditional_answer	Object	Examined+	Support, rejection, non-decisive conditions, scope, non-claims.
status_assignment	Object	Always	Primary status, rationale, transition requirements, two-sided check.
review_record	Object	Always	Version-specific passes, findings, and separate review results.
boundary_record	Object	Always	Operational-content classification and non-suppression rule.
lineage	Object	Always	Predecessors, supersession, related documents, changes.
machine_validation	Object	Always	Schema and invariant results, validator, errors, warnings.
publication_lock	Object	Always	False/null in drafts; hashes and canonical record when locked.
branch_scope_assessment	Object	0.3 records	Distinguish selected-pair, branch-family, and mixed/sequential claim scope; require family support or claim narrowing.
human_review_taxonomy	Object	0.3 records	Separate authorial human review, AI-informed understanding, external independent review, and external peer review.
administrative_successor	Object	0.3 records	Record zero-drift promotion, predecessor, fresh validation, review attribution, and authorization.

Appendix C. Accounting-Ledger Template

C.1 Core charge routing

ID	Branch	Outcome	Label	Causal source	Parent pathway	Directness	Sign
L-001	B-REF	Y-COST	Verification burden	Direct verification process	None	DIRECT	COST

ID	Branch	Outcome	Label	Causal source	Parent pathway	Directness	Sign
L-002	B-ALT	Y-COST	Enforcement burden	Behavioral resistance response	Pathway-P1	DERIVED	COST
L-003	B-ALT	Y-INFO	Information degradation	Suppressed corrective signals	Pathway-P1	INTERACTION	COST

C.2 Interaction and aggregation control

ID	Value status	Overlap group	Interaction class	Aggregation authorization	Independent-sum reason
L-001	SYMBOLIC	None	INDEPENDENT	AUTHORIZED	Independent addition justified by separate causal source.
L-002	UNIDENTIFIED	OG-1	CAUSALLY_DERIVED	CONDITIONAL	Requires joint treatment with resistance cost.
L-003	BOUNDED	OG-1	NONLINEAR_INTERACTION	NOT_AUTHORIZED	Reported separately; not independently summed.

Production records store the complete ledger fields in JSON and may export a separate CSV for detailed auditing. The residual interaction is represented once at the accounting-ledger object level and must name the ledger IDs it excludes.

Appendix D. Cycle-Resolution Decision Sequence

1. Build the directed dependency graph using PREREQUISITE and DEPENDS_ON relations.
2. Compute strongly connected components. A component of one node with no self-loop is acyclic.
3. For each cyclic component, classify the cycle as definitional, parametric, dynamic, strategic, or mixed.
4. Write the simultaneous system or best-response correspondence explicitly.
5. Select a resolver only after stating the conditions under which it is meaningful.
6. Record existence and uniqueness separately. Do not infer uniqueness from one numerical solution.
7. For numerical methods, record initialization, tolerance, stopping rule, and convergence status.
8. Search for alternative roots, equilibria, trajectories, or set-valued solutions when multiplicity is plausible.
9. Test whether the branch ordering is invariant across admissible solutions.
10. If no justified solution or robust ordering exists, preserve the cycle as UNRESOLVED, MODEL_DEPENDENT, CONFLICTING_RESULTS, or SIGN_INDETERMINATE as appropriate.

Strategic-cycle caution
A game-theoretic equilibrium is a conditional solution concept, not a prediction that real agents will play it. The record must preserve information, rationality, and equilibrium-selection assumptions.

Appendix E. Review and Pre-Lock Checklist

Review area	Pass condition
Identity	Question ID, examination ID, versions, canonical wording, and family match the registry.
Scope	Included/excluded meanings, unit, affected agents, horizon, and prohibited detail are explicit.
Branches	At least two admissible branches; comparator is not silently privileged.

Review area	Pass condition
Variables	Symbols, roles, units, values, bounds, and sources are complete.
Assumptions	Classes, necessity, relaxation tests, and equation links are complete.
Evidence	Empirical sources, derivations, illustrations, and judgments are separated.
Outcomes	Vector dimensions and directions are explicit; scalarization is justified or refused.
Accounting	No duplicate charge; correlations, overlap groups, and residual interaction audited.
Cycles	Strongly connected components and solution methods are disclosed.
Mathematics	Equations, sign convention, solution method, existence, uniqueness, and limitations reviewed.
Identification	Controlling empirical and normative limits are named.
Sensitivity	Preferred, reversal, threshold, horizon, weight, and residual regions tested where applicable.
Counterexamples	At least one serious adversarial construction attempted.
Answer	Conditional answer matches the derivation and explicitly states non-claims.
Status	Primary status, secondary flags, transition requirements, and two-sided openness reviewed.
Boundary	Restricted implementation detail excluded without suppressing the abstract result.
JSON	Schema validation and invariants pass; all referenced IDs resolve.
PDF/JSON	Exact-version identifiers, status, equations, thresholds, and conclusion agree.
Visual	Every page inspected; no identifier, table, equation, header, footer, or glyph defects.
Lock	Final lock record, hashes, canonical URI, retrieval verification, and successor rule complete.
Claim scope	Selected pair versus branch-family scope is declared; unsupported family claims are narrowed.
Review disposition	PASS WITH NON-BLOCKING NOTES is encoded directly rather than hidden in narrative.
Administrative successor	Zero-drift audit, predecessor attribution, fresh validation, and authorization are complete.
Human review taxonomy	Authorial human review and external independent peer review are separately and accurately recorded.